



# SAA Win Probability Model

Shelby Young and Eric Merdian

Data Science Capstone, Centre College, Danville, Kentucky



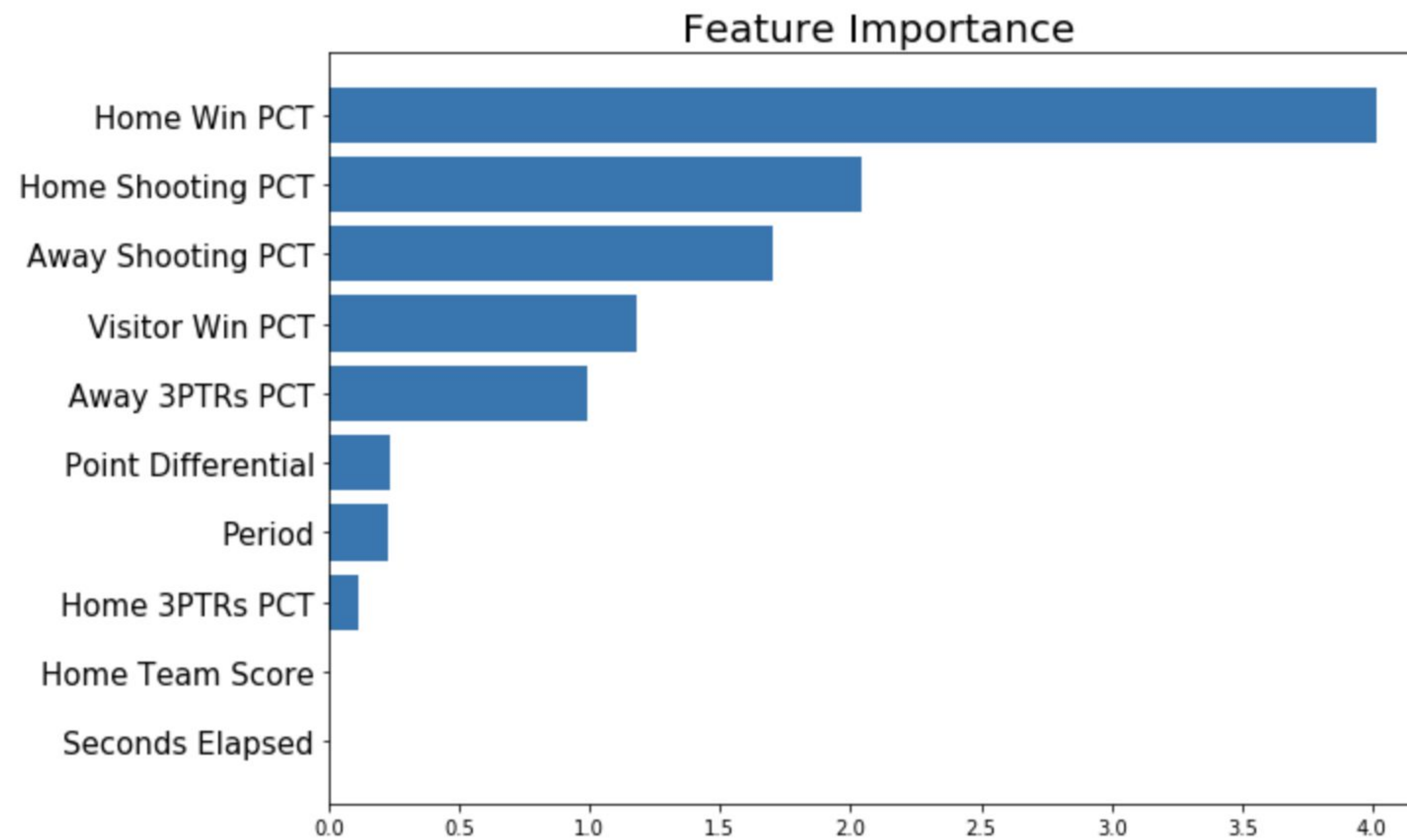
## Objective

Create a win probability model for Centre Sports Analytics to use based on data from the past 10 years in 13 different men's and women's sports

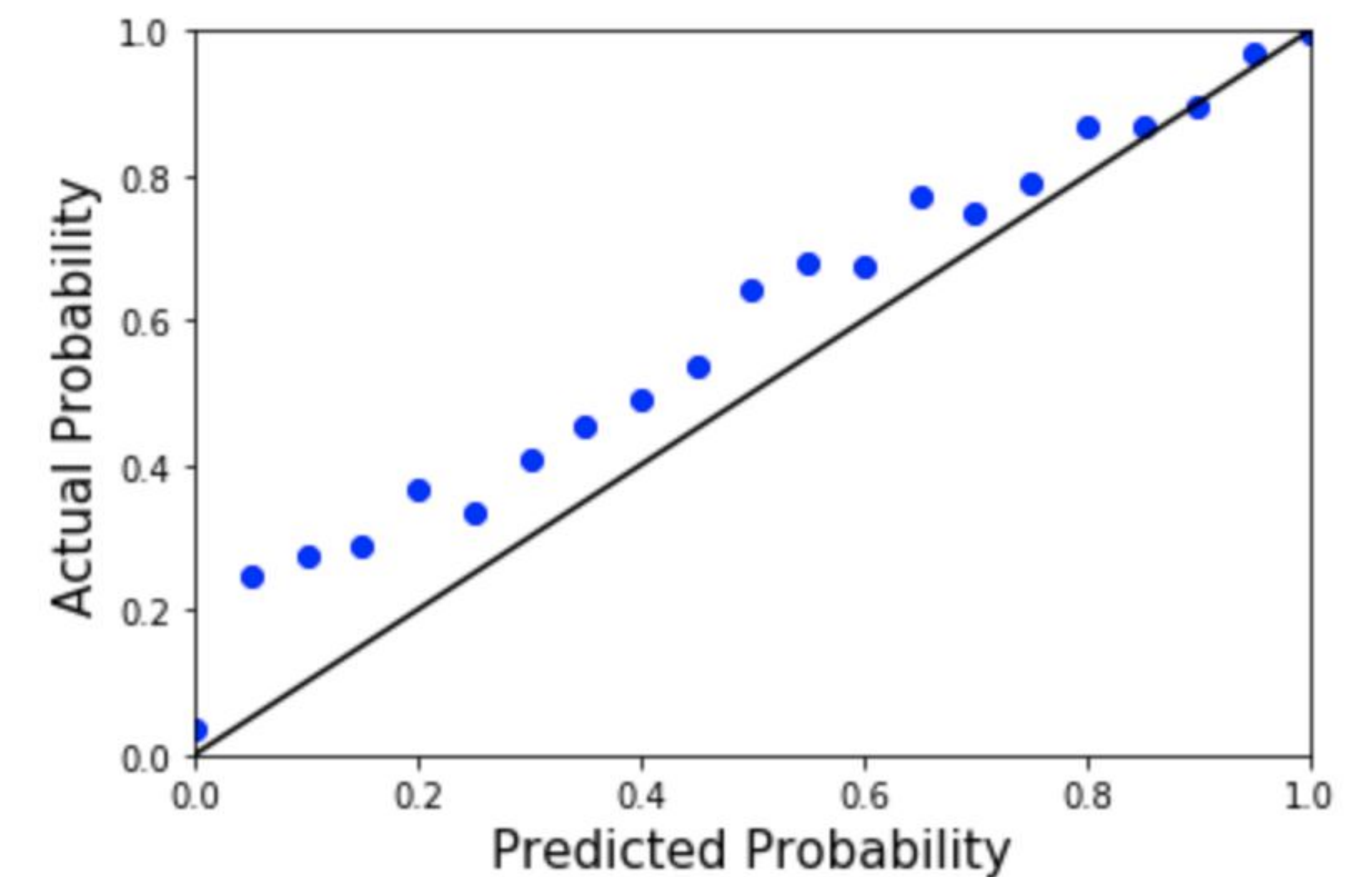
## Gathered Data

| SEWANEE |                                             |        |  |        | CENTRE-M                                 |  |  |  |  |
|---------|---------------------------------------------|--------|--|--------|------------------------------------------|--|--|--|--|
| 19:33   |                                             |        |  |        | FOUL by BATES,JACOB                      |  |  |  |  |
| 19:33   | GOOD FT by WARLICK,JORDAN                   | 1 (+1) |  | 0      |                                          |  |  |  |  |
| 19:33   | MISS FT by WARLICK,JORDAN                   |        |  |        |                                          |  |  |  |  |
| --      |                                             |        |  |        | REBOUND DEF by WALKER,ART                |  |  |  |  |
| 19:13   |                                             | 1      |  | 2 (+1) | GOOD LAYUP by RAKES,BAILEY(in the paint) |  |  |  |  |
| 18:52   | TURNOVER by BUESCHER,HUNTER                 |        |  |        |                                          |  |  |  |  |
| 18:50   |                                             |        |  |        | STEAL by RAKES,BAILEY                    |  |  |  |  |
| 18:47   |                                             |        |  |        | MISS 3PTR by WALKER,ART                  |  |  |  |  |
| --      |                                             |        |  |        | REBOUND OFF by TEAM                      |  |  |  |  |
| 18:41   |                                             |        |  |        | TURNOVER by AYERS,PERRY                  |  |  |  |  |
| 18:19   | MISS 3PTR by SMITH,LUKE                     |        |  |        |                                          |  |  |  |  |
| --      | REBOUND OFF by TEAM                         |        |  |        |                                          |  |  |  |  |
| 18:18   |                                             |        |  |        | FOUL by SINE,TUCKER                      |  |  |  |  |
| 18:13   | GOOD 3PTR by WARLICK,JORDAN                 | 4 (+2) |  | 2      |                                          |  |  |  |  |
| 17:45   |                                             | 4      |  | 4      | GOOD LAYUP by BATES,JACOB(in the paint)  |  |  |  |  |
| 17:15   | GOOD LAYUP by BUESCHER,HUNTER(in the paint) | 6 (+2) |  | 4      |                                          |  |  |  |  |
| 17:02   |                                             |        |  |        | MISS LAYUP by AYERS,PERRY                |  |  |  |  |

## The Model and its Variables



## Results and Analysis



## Application and Future Work

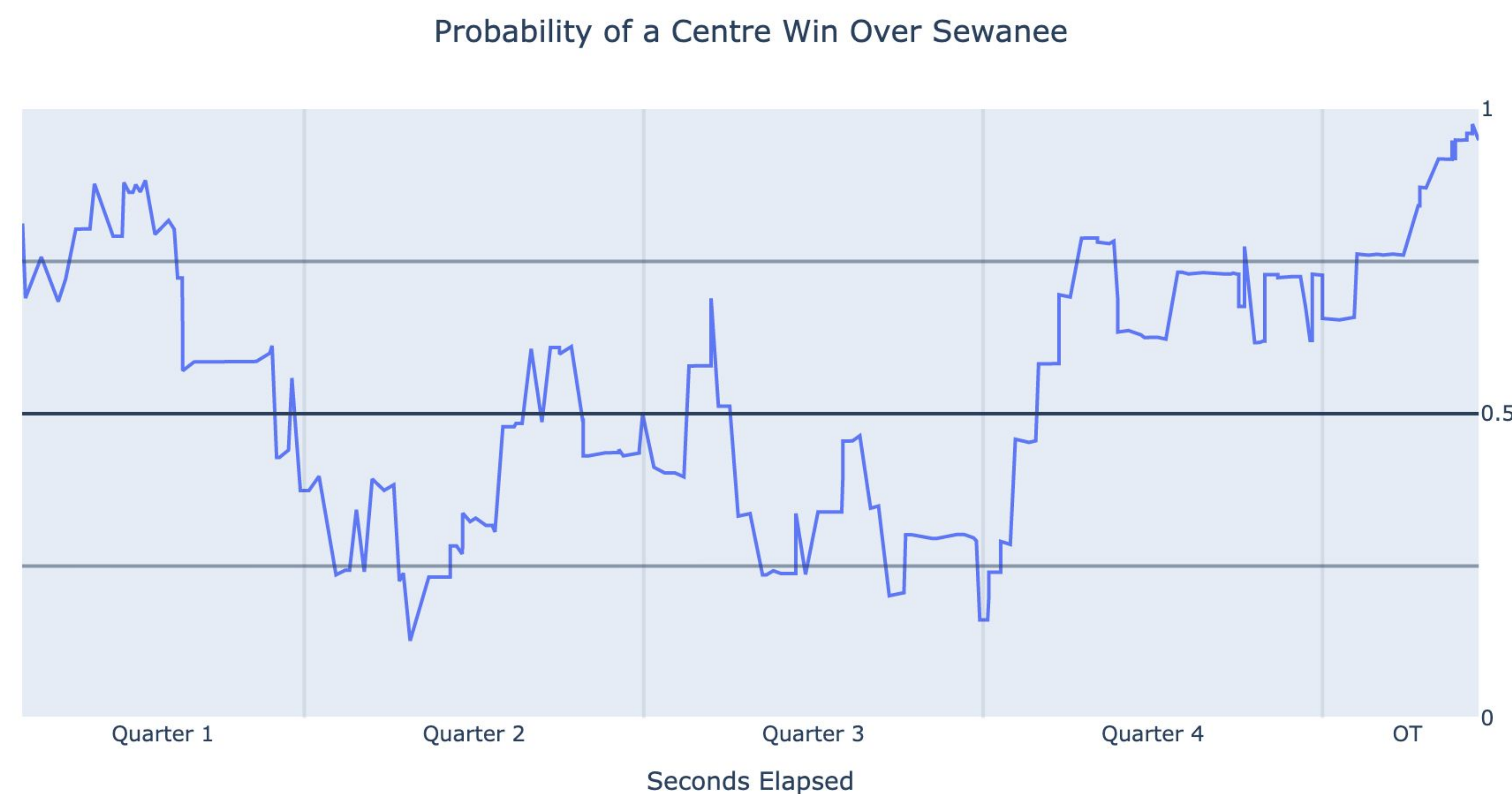
Looking into the future, we would like to expand our model with the use of data from more years as well as going beyond modeling Basketball and Lacrosse.

With this model, we would be interested in pitching this model to Sidearm Sports, the builder of the SAA member colleges websites to include as an in-game metric.

## Acknowledgements and References

We would like to thank the colleges of the SAA for making data available to use, as well as the athletes who created the data for us. We would also like to thank Drs. Heath and Allen for guiding us through this project and providing help along the way.

## Example Men's Basketball Model







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| SEWANEE |                             |        |                                                                                       |        | CENTRE-M                                 |
|---------|-----------------------------|--------|---------------------------------------------------------------------------------------|--------|------------------------------------------|
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| 19:13   |                             | 1      |    | 2 (+1) | GOOD LAYUP by RAKES,BAILEY(in the paint) |
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| 18:47   |                             |        |                                                                                       |        | MISS 3PTR by WALKER,ART                  |
| --      |                             |        |                                                                                       |        | REBOUND OFF by TEAM                      |
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| 18:19   | MISS 3PTR by SMITH,LUKE     |        |                                                                                       |        |                                          |
| --      | REBOUND OFF by TEAM         |        |                                                                                       |        |                                          |
| 18:18   |                             |        |                                                                                       |        | FOUL by SINE,TUCKER                      |
| 18:13   | GOOD 3PTR by WARLICK,JORDAN | 4 (+2) |  | 2      |                                          |
| 17:45   |                             | 4      |  | 4      | GOOD LAYUP by BATES,JACOB(in the paint)  |